



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

[Project Proposal]

Project Title: AI-Based Smart Quiz System.

Course Code: CSE 4204

Course Title: Mobile Computing Lab

Submitted By:

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GitHub Repository: <https://github.com/sudipto224/AI-based-Smart-Quiz-System>

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1. Project Title

AI-based Smart Quiz System.

2. Problem Statement

2.1 The Problem Currently Exists

Creating high-quality multiple-choice questions is a demanding and time-consuming process. Educators and administrators must manually compose each question, formulate four plausible answer options, designate the correct answer, and write a meaningful explanation all while ensuring factual accuracy and appropriate difficulty. As courses evolve and question banks grow, this burden compounds, leaving little time for instructors to focus on actual teaching.

Simultaneously, online quiz environments are highly susceptible to academic dishonesty. Without physical supervision, students can easily switch to other browser tabs to consult external resources, or attempt to game the system by submitting answers within impossibly short time spans. These behaviours undermine the validity of scores, distort leaderboard rankings, and erode the credibility of digital assessments.

2.2 The Problem Importance

Reliable and efficient assessment is fundamental to any educational programme. Teachers and administrators need tools that reduce the overhead of content creation so they can allocate more time to instruction and feedback. At the same time, assessment integrity is critical: if students can cheat with impunity, exam results lose their meaning as indicators of genuine learning, which affects grading fairness, course certification, and institutional reputation.

2.3 The Existing Solutions Are Insufficient

Popular quiz and e-assessment platforms such as Google Forms, Kahoot, and Typeform provide basic question authoring and delivery features but fall short in several key respects. They do not offer any form of AI-driven question generation, meaning every question must still be written by a human from scratch. They provide no real-time monitoring for cheating

behaviour such as tab switching, nor do they flag statistically suspicious submission speeds. Administrators are therefore left without actionable data to identify potentially dishonest attempts, and with no automated means to rapidly expand or refresh their question banks.

2.4 The Relevance of Artificial Intelligence

Large language model APIs such as Google Gemini and OpenAI GPT are capable of generating factually accurate, well-structured multiple-choice questions on virtually any topic within seconds. By integrating such an API into the admin panel, the system can dramatically reduce the time and expertise required to build a diverse question bank. Furthermore, AI-adjacent data analysis specifically, computing statistical thresholds on time-per-question ratios provides an automated, objective method for identifying suspiciously fast submissions, replacing the need for manual review of every attempt.

3. Objectives

The primary objectives of the AI-based Smart Quiz System are as follows:

1. To provide a secure, role-based authentication system that distinguishes between standard users and administrators, directing each to the appropriate interface upon login.
2. To deliver an engaging and time-bound quiz experience across three difficulty levels Easy, Medium, and Hard each with a configurable per-question countdown timer, a visual low-time warning, and automatic submission on timeout.
3. To present detailed post-quiz feedback, including the student's total score, a list of incorrectly answered questions, the correct answers, and database-stored explanations, enabling self-directed learning.
4. To maintain a competitive leaderboard that ranks all participants by their best score and, as a tiebreaker, by their lowest completion time, promoting healthy academic engagement.

5. To empower administrators with a dedicated panel for adding, viewing, and deleting quiz questions across all difficulty levels, with each question storing its level, text, four options, correct answer, and explanation.
6. To integrate an AI API (Google Gemini or OpenAI GPT) that enables administrators to automatically generate complete multiple-choice questions including options, correct answer, and explanation from a simple topic prompt, reducing content-creation overhead significantly.
7. To implement a smart cheating detection mechanism that monitors student behaviour during quizzes in real time, detecting browser tab switches and flagging statistically suspicious submission speeds so that administrators can review and act on potentially dishonest attempts.

4. Proposed Features

4.1 Authentication and User Management

The proposed system will implement a secure, role-based authentication mechanism using Laravel Breeze as the foundational scaffold, which will be customised to meet the specific requirements of this application. Upon successful registration, the system will redirect the user to the login page and display a confirmation message; automatic login will not occur at this stage. During the login process, the system will examine the `is_admin` flag stored in the users table. Based on the value of this flag, authenticated users will be directed to the appropriate interface: those with `is_admin = 1` will be routed to the dedicated administration panel, while standard users will be redirected to the public home page. All protected routes will be guarded by custom middleware to prevent unauthorised access, ensuring that only legitimate users can reach the intended sections of the platform.

4.2 Quiz Taking and Timer

Users may begin a quiz session by selecting a difficulty level from the Start Quizzing modal on the home page or from the Select Level navigation dropdown. Three levels are available: Easy, Medium, and Hard. Each question is presented individually with four radio-button

answer options. A per-question countdown timer defaulting to 30 seconds and configurable per level governs response time. When 10 or fewer seconds remain, the timer border pulses red to alert the student. If no answer is selected before the timer expires, the system auto-submits the current question and advances to the next.

4.3 Result Page and Explanation

Upon completing all questions in a session, the student is presented with a result page that displays their total score. Below the score, every incorrectly answered question is listed alongside the student's selected option, the correct answer, and the explanation stored in the questions table. Two action buttons allow the student to retake the same level or return to the level-selection modal to choose a different difficulty.

4.4 Leaderboard

The leaderboard aggregates data from the quiz_attempts table and ranks all registered users by their best (highest) score. Where scores are equal, the user who achieved that score in the shortest time is ranked higher. The leaderboard table displays each participant's rank, name, best score, and best time in seconds.

4.5 Admin Panel -Question Management

A dedicated administration panel is accessible exclusively to users whose is_admin column is set to 1. The panel provides a question-addition form capturing the difficulty level (Easy, Medium, or Hard), the question text, four answer options, an explanation, and the correct answer selected from a dynamically populated dropdown that mirrors the entered options. All questions in the database are listed below the form, each accompanied by a Delete button, enabling straightforward removal of outdated or incorrect content. All question data is stored in the questions table.

4.6 AI-Powered Question Generation

The admin panel will include a Generate Question with AI section. The administrator enters a topic of their choice for example, JavaScript Closures or Newton Laws of Motion and optionally selects a target difficulty level. On submission, the Laravel backend constructs a structured prompt and dispatches it to the configured AI API (Google Gemini or OpenAI

GPT). The API responds with a JSON object containing the question text, four answer options, the correct answer, and a concise explanation. The backend parses this response and inserts the data directly into the questions table. A confirmation message is displayed, and the new question immediately appears in the question list. The administrator may review and edit the generated content at any time using the standard question management interface, ensuring that human oversight is retained over all AI-generated material.

4.7 Cheating Detection

Two complementary mechanisms protect the integrity of each quiz session.

Tab Switch Detection: As soon as a quiz session begins, a JavaScript event listener is attached to the document visibilitychange event. If the quiz page becomes hidden — because the user switches to another browser tab, minimises the window, or opens a new application the system immediately displays a prominent warning message on screen. Tab-switch events may also be logged server-side with a timestamp and the associated quiz attempt identifier, creating an auditable record for administrative review without immediately terminating the session.

Early Submission Detection: The quiz start time is recorded at session initialisation. When the student submits the quiz, the server calculates the total time elapsed and divides it by the number of questions to obtain the average time per question. If this average falls below a configurable threshold set by default to 5 seconds per question the corresponding record in the quiz_attempts table is flagged by setting a new is_suspicious column to true. A dedicated view in the admin panel lists all flagged attempts with the student name, quiz level, score achieved, and total time taken, enabling the administrator to make an informed judgement about each submission.

The is_suspicious column will be added to the quiz_attempts table via a new Laravel database migration.

5. Technology Stack

Frontend :

- HTML5 : semantic markup for all views including homepage, quiz page, result page, leaderboard, and admin panel.
- Custom CSS : bespoke styling covering the hero section, gradient background, floating animated shapes, modal overlay, and responsive layout. No CSS framework such as Tailwind is used.
- Vanilla JavaScript : countdown timer logic, red-pulse low-time warning, auto-submit on timeout, dynamic correct-answer dropdown population in the admin form, tab-switch detection via the visibilitychange event, and modal control.

Backend :

- Laravel 12 (PHP) : MVC application framework providing routing, middleware, Eloquent ORM, database migrations, and all server-side business logic including AI API communication and suspicious-attempt flagging.

Database :

- PostgreSQL : relational database management system storing all application data across the users, questions, quiz_attempts, sessions, password_reset_tokens, and migrations tables.

AI Integration :

- Google Gemini API or OpenAI API large language model API called from the Laravel backend to generate multiple-choice questions in structured JSON format from administrator-supplied topic prompts.

Authentication :

- Laravel Breeze lightweight authentication scaffolding customised to support role-based post-login redirection and registration-confirmation messaging.

Version Control :

- Git : source code versioning and history management throughout the development lifecycle.

6. Expected Outcome**6.1 Benefits for Students**

Students will gain access to a structured, gamified learning environment with progressively challenging difficulty tiers. The immediate post-quiz feedback showing which questions were answered incorrectly, the correct answers, and clear explanations transforms each quiz session from a passive test into an active learning experience that reinforces correct understanding and highlights knowledge gaps. The competitive leaderboard will motivate regular participation and improvement. The consistent timer-based format will also help students develop time-management skills applicable to formal examinations. Because cheating detection mechanisms ensure that leaderboard rankings reflect genuine performance, students who study honestly will benefit from a fair and credible assessment environment.

6.2 Improvements to Existing Workflow

Administrators will experience a substantial reduction in the time and effort required to maintain and expand the question bank. AI-powered question generation means that a diverse set of new questions on any topic can be produced in seconds rather than hours. This allows the question bank to be refreshed frequently, reducing the risk of students becoming familiar with recurring questions. The suspicious-attempt review feature eliminates the need for manual inspection of every quiz submission: the system automatically surfaces only those attempts that exhibit statistically anomalous behaviour, enabling administrators to focus their review effort where it matters most.

6.3 Benefits of AI Integration

The integration of a large language model API fundamentally changes the economics of question authoring. Educators who lack the time or inclination to compose detailed multiple-choice questions can delegate the initial drafting to the AI and retain editorial control by reviewing and editing before saving. This human-in-the-loop approach ensures content quality while dramatically lowering the barrier to producing a rich and varied question bank. On the integrity side, the combination of real-time browser event monitoring and server-side statistical analysis creates a scalable detection framework that grows with the user base without requiring dedicated human invigilators. Together, these AI-powered capabilities position the AI-based Smart Quiz System as a more efficient, trustworthy, and future-ready assessment platform than existing generic solutions.